SE200 PCP Drive

Version 1.0

Operations Manual

SOFTWARE VERSION 1.9

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Keypad (HMI)

The main screen of the SE200's HMI includes several buttons and indicators to provide feedback and control of the VFD. An example of the main screen is shown in the figure below.

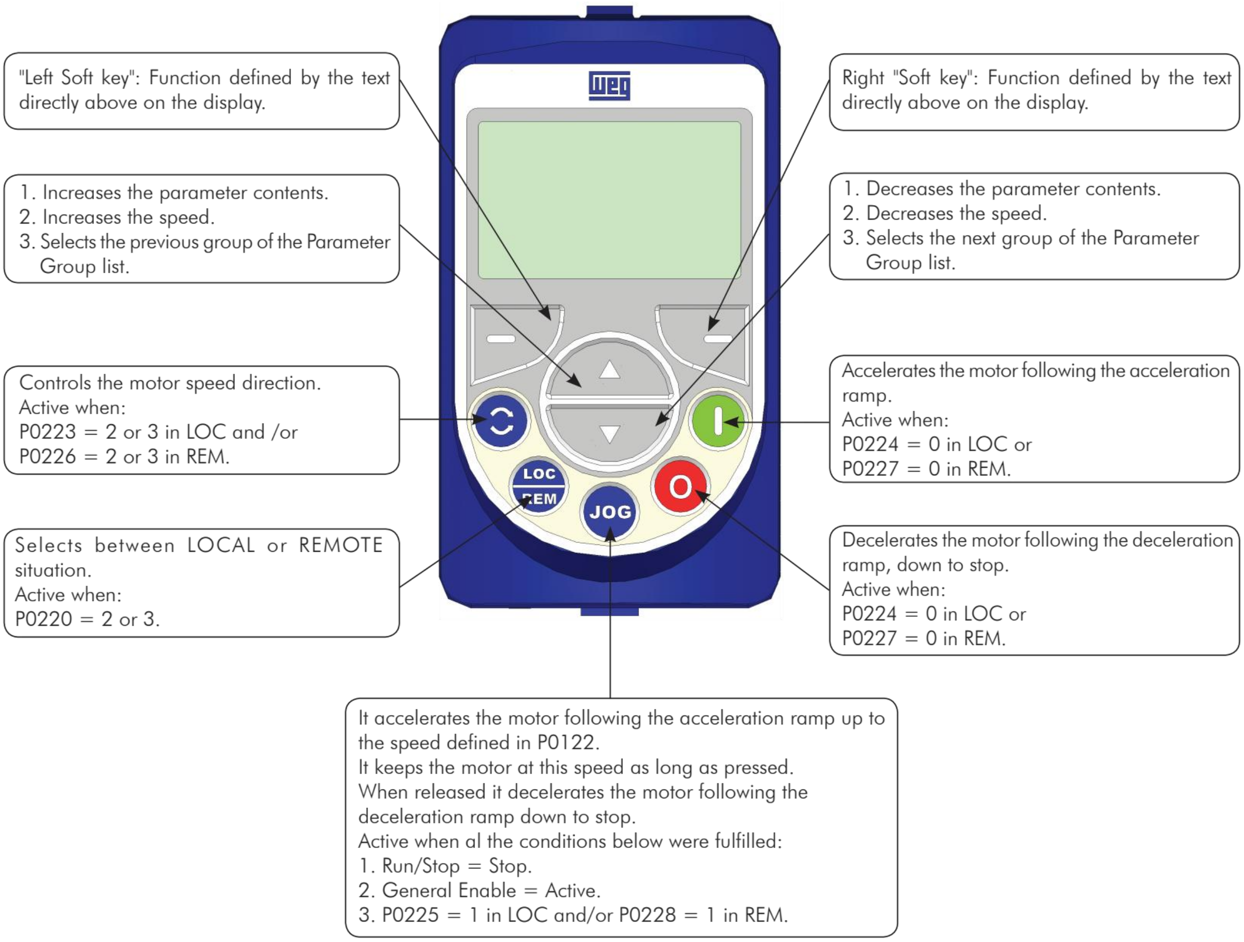


Figure 1 Keypad

Power and Motor Terminations

The SE200 VFD supports both single-phase and three-phase applications. When using it in a single-phase setup, follow these key guidelines to ensure proper operation:

1. **Power Input Connections**

- Connect **single-phase** power to A/L1 **(Red)** and B/L2 terminals **(Black)**. Connect **three-phase** power to A/L1 **(Red)**, B/L2 terminals **(Black)** and C/L3 terminals **(Blue).**
  - These terminals supply power to the DC bus of the CFW11.

2. **Motor Wiring**
  - Terminate the motor leads at the U, V, and W terminals, located at the bottom right of the VFD.

3. **Sizing Requirement**
  - When using the SE200 in single-phase mode, select a VFD rated at twice the intended motor load.
  - This ensures sufficient capacity and reliable performance.

4. **Parameter Configuration**
  - Set P0357 = 0 sec to bypass the phase loss fault (this is required for operation).

Following these steps will help ensure efficient and reliable operation of the SE200 VFD in a single-phase configuration

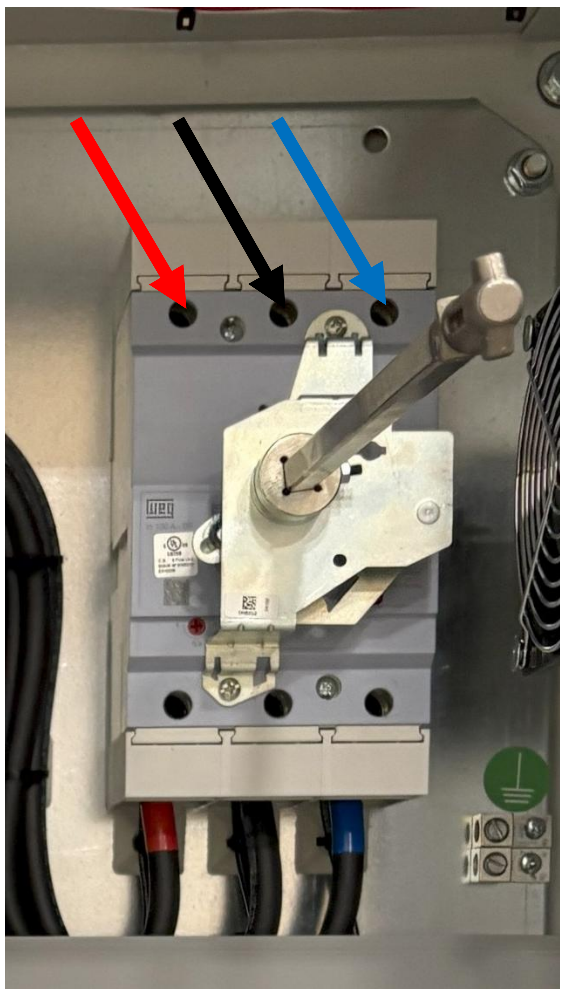


Figure 2 Main Breaker

Control Wiring

For external shutdowns, such as Presco, Vibration Switches the control wiring can be connected as outlined below.

1. **Jumper Removal**
   - Remove jumpers only for the specific shutdowns that are required.
2. **Alarm Indications**
   - The following alarms will appear on the HMI for corresponding shutdowns:

- A750 – High Torque Shutdown
- A751 – Backspin Timer
- A752 – Shutdown #1 (Auto Restart can be enabled)
- A753 – Shutdown #2 (Does not have auto restart capability)
- A754 – Shutdown #3 (Does not have auto restart capability)
- A755 – Auto Restart Wait Time Delay

3. **Additional Information**
   - Refer to the wiring diagram for detailed instructions and additional guidance.

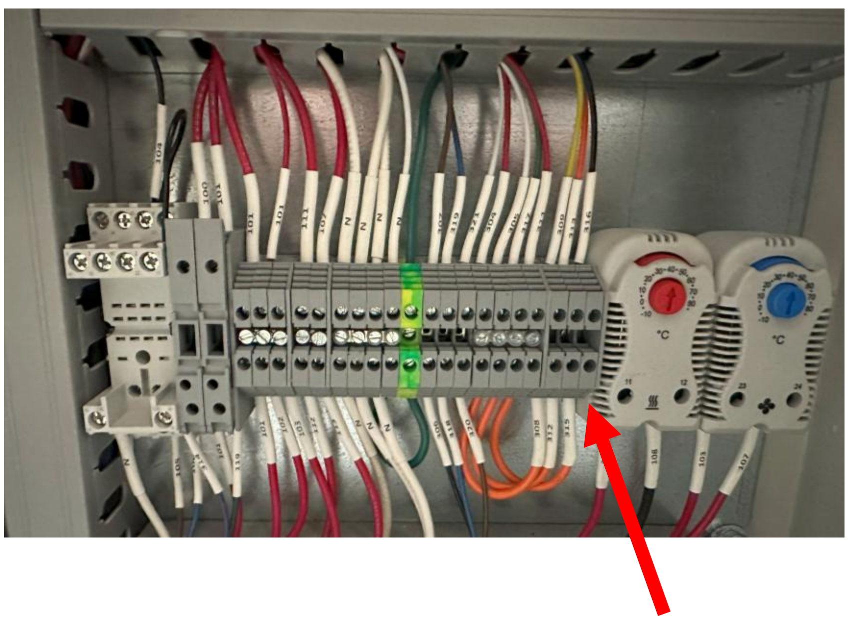


Figure 3 Basic terminal strip layout

Initial Power Up

When powering up the **SE200 PCP** for the first time, the field technician must configure the device using the **Field Technician Parameters** section outlined in our startup guide. These parameters are detailed below, and a full copy of the guide is included with every **SE200** unit. **Parameter steps #1-#30 are set in SEEL factory and do not need to be set in the field unless otherwise told by SEEL Representative, or in other parts of this manual.**

Steps to Access and Configure Parameters:

1. **Access the Parameter Section**
    - Press **Menu** > **Select All Parameters**.
    - Use the **Up** and **Down** arrows to navigate to parameter **P0202**.
2. **Complete the Oriented Startup Menu**
    - Starting at Step 1, the SE200 VFD will guide you through the oriented startup steps **(Steps 31–38).**
    - After completing the auto-tune process, the device will return to the main VFD screen.
3. **Continue Programming**
    - Return to **Menu** > **All Parameters** to proceed with the programming procedure **(Steps 39–54).**

PARAMETERS TO BE SET BY FIELD PERSONNEL (Steps #1-42)					
Step#	Parameter #	Parameter Description	Unit	Default Value	Set Value
1	P0000	Access to Parameters	0-9999	0000	0005
2	P0100	Acceleration Time	Sec	20.0	5.0
3	P0101	Deceleration Time	Sec	20.0	5.0
4	P0132	Max Over Speed Level	0-100%	10%	50%
5	P0205	Read Parameter Selection #1	SoftPLC P1014	2	25
6	P0207	Read Parameter Selection #2	SoftPLC P1019	3	30
7	P0217	Zero Speed Disable	Disable Magnetization Current	0	1
8	P0220	LOC/REM Selection Source	LR Key LOC	2	2
9	P0221	LOC Reference Select	AI1	0	1
10	P0222	Remote Reference Select	Serial/USB	1	9
11	P0223	LOC FWD/REV	Always FWD	2	0

12	P0224	LOC Run/Stop Selection	SoftPLC	0	5
13	P0225	LOC Jog Selection	Disable	1	0
14	P0226	REM FWD/REV	Always FWD	4	0
15	P0227	REM Run/Stop Selection	Serial/USB	1	2
16	P0228	REM Jog Selection	Serial/USB	2	3
17	P0229	Stop Mode Selection	Coast to Stop	0	1
18	P0233	AI1 Signal Type	0-10vdc (0) or 4-20mA (1)	0	0
19	P0238	AI2 Signal Type	0-10vdc (0) or 4-20mA (1)	0	Either 0 or 1
20	P0263	DI1 Function	PLC Use (Backspin Bypass)	1	21
21	P0264	DI2 Function	PLC Use	8	21
22	P0265	DI3 Function	PLC Use	0	21
23	P0266	DI4 Function	PLC Use	0	21
24	P0267	DI5 Function	PLC Use (Shutdown Bypass)	10	21
25	P0268	DI6 Function	General Enable	14	2
26	P0275	DO1 Function (RL1)	Soft PLC (General Enable)	13	28
27	P0276	DO2 Function (RL2)	Soft PLC (Red Alarm Beacon)	2	28
28	P0277	DO3 Function (RL3)	Run (Green Run Beacon)	1	11
29	P0320	Fly Start/Ride-Through	Ride-Through	0	3
30	P0353	IGBT's / Air Overtemp Config	Under Temp Disable	0	4
31	P0202	Type of Control	Sensorless Vector	0	3
32	P0398	Motor Service Factor	As per motor nameplate		Motor SF
33	P0400	Motor Rated Voltage	As per motor nameplate		Motor Voltage
34	P0401	Motor Rated Current	As per motor nameplate		Motor Current
35	P0402	Motor Rated Speed	As per motor nameplate		Motor RPM
36	P0403	Motor Rated Frequency	As per motor nameplate		Motor Hz
37	P0404	Motor Rated Power	As per motor nameplate		Motor HP/KW
38	P0408	Run Self Tuning	No Rotation (if "Close General Enable" is displayed ensure on/off switch is on)	0	1

39	P0156	Overload Current 100% Speed	0.1 to 1.5 x Inom-ND Example: Motor FLA – 13.2A If SF is 1.0, P0156 should be set to 15.2A If SF is 1.15, P0156 should be set to 16.5A	1.05 x Inom-ND	As per motor service factor, adjust if required (SF=1.0 – 1.15 x Inom-ND) (SF=1.15 – 1.25 x Inom-ND)
40	P0157	Overload Current 50% Speed	0.1 to 1.5 x Inom-ND DO NOT ADJUST UNLESS REQUIRED	0.9 x Inom-ND	As per motor service factor, adjust if required
41	P0158	Overload Current 5% Speed	0.1 to 1.5 x Inom-ND DO NOT ADJUST UNLESS REQUIRED	0.65 x Inom-ND	As per motor service factor, adjust if required
42	P0159	Motor Thermal Class	Class 5,10,15,20,25,30,35,40,45	1 – Class 10	3 – Class 20
43	P0169	Max + Torque Current	0-350% (Adjust if required)	125%	150%
44	P0170	Max – Torque Current	0-350% (Adjust if required)	125%	150%
45	P0296	Line Rated Voltage	0-6 (Can be adjusted if overvoltage or undervoltage trips occur)	4- (480)	Set as per input voltage
46	P0185	DC Link Regulation Level	Adjustable Voltage Range as per input voltage	800V	770V
47	P0186	DC Link Proportional Gain	0-63.9	18	40
48	P0321	DC Link Power Loss	308-616V	505V	Adjust if required

49	P0322	DC Link Ride-Through	308-616V	490V	Adjust if required
50	P0323	DC Link Power Back	308-616V	535V	Adjust if required
51	P0325	Ride-Through P Gain	0.0-63.9	22.8	10.0
52	P0340	Auto Reset Time (VFD Faults)	0-3600s	0	Set to desired reset time
53	P0357	Line Phase Loss Time	0-60sec	3 sec	Set to 0 seconds
54	P1001	Soft PLC Command	Ensure this is set to “Run Program”	0	1

Table 1 Startup guide

Equipment Information

Steps 55-73: Equipment Information and Configuration

To program steps 55-73, you must gather specific pumpjack details. The technician will need to obtain the following:

1. **Sheave Ratio ex 4:1, 6:3. This parameter is calculated by dividing the drive sheave by the motor sheave**

Programming Steps Breakdown:

- **Steps 55-73:** Highlighted in **blue**, these steps are required for the operation of the SE200.

Setting Parameters:

When setting the parameters for the SE200 PCP the following must be taken into consideration.

- **P1012- Backspin Lockout Time –** This is the time required for the pump to complete its backspin time in minutes. A typical setpoint is around 30min but can vary depending on application.
- **P1015/1016-** Min/Max Polish Rod Speed – These parameters are well dependent and can vary between application.
- **P1018-** Torque Shutdown Setpoint – This is the setpoint for high torque shutdown this should be set conservatively as it is intended to protect downhole equipment.

Additional Notes:

- **Ensure rotation is correct before connecting the belts on the sheave**

PARAMETERS TO BE SET BY FIELD PERSONNEL (Steps #55-72)					
55	P1010	Sheave Ratio	Calculate the Sheave Ratio by dividing the diameter of the drive sheave / motor sheave	0	Calculated Sheave Ratio
56	P1012	Back Spin Lockout Time	0-2000 mins	0	Desired Back Spin Lockout Time
57	P1013	Back Spin Time Elapsed	Read Only Parameter	n/a	n/a
58	P1014	Polish Rod Speed	Read Only Parameter	n/a	n/a
54	P1001	Soft PLC Command	Ensure this is set to "Run Program"	0	1
59	P1015	Minimum Polish Rod Speed	0-1800	0	Desired Minimum Polish Rod Speed
60	P1016	Maximum Polish Rod Speed	0-1800	0	Desired Maximum Polish Rod Speed
61	P1017	Torque Shutdown Debounce Setpoint	0-100	0	Desired Torque Shutdown Debounce Setpoint
62	P1018	Torque Shutdown Setpoint (FT-LB)	0-3000	0	Desired Torque Shutdown Setpoint
63	P1019	Polish Rod Torque	Read Only Parameter	n/a	n/a
64	P1020	DI2 Shutdown Auto Restart Time (Sec)	0-32767	0	Desired Auto Restart Wait Time
65	P1021	DI2 Shutdown Auto Restart Enable	0-1	0	Set to 1 if auto restart is desired
66	P1022	DI2 Shutdown Trips Allowed Before Lockout	0-10	0	Desired amount of presco trips
67	P1023	Power loss Restart Enable (allows for automatic restart after power loss)	0-1	0	Set to 1 if auto restart is desired after power loss

		Enabling P1023 (Powerloss Restart Enable) allows the VFD to restart automatically after a power loss or fault without requiring the Green Button to be pressed			
68	P1024	Power loss Restart Time (Sec)	0-32767	0	Set to desired restart wait time after a power loss
69	P1025	Shutdown Bypass Time (Sec) Time allowed to test shutdowns #1,#2,#3 while VFD stays running (if equipped the red beacon will illuminate when shutdown is tripped)	0-1800	0	Set to desired time the bypass will be activated after door mounted pushbutton is pressed
70	P1026	Shutdown #1 Debounce Timer (Sec)	Set to desired shutdown debounce time up to 5sec	0	Suggest leave at 0 unless problems persist
71	P1027	Shutdown #2 Debounce Timer (Sec)	Set to desired shutdown debounce time up to 5sec	0	Suggest leave at 0 unless problems persist
72	P1028	Shutdown #3 Debounce Timer (Sec)	Set to desired shutdown debounce time up to 5sec	0	Suggest leave at 0 unless problems persist
73	P1029	Shutdown Bypass Time Elapsed (Sec)	Displays the shutdown bypass time counter	Read only parameter	n/a

Table 2 Pumpjack parameters

Troubleshooting

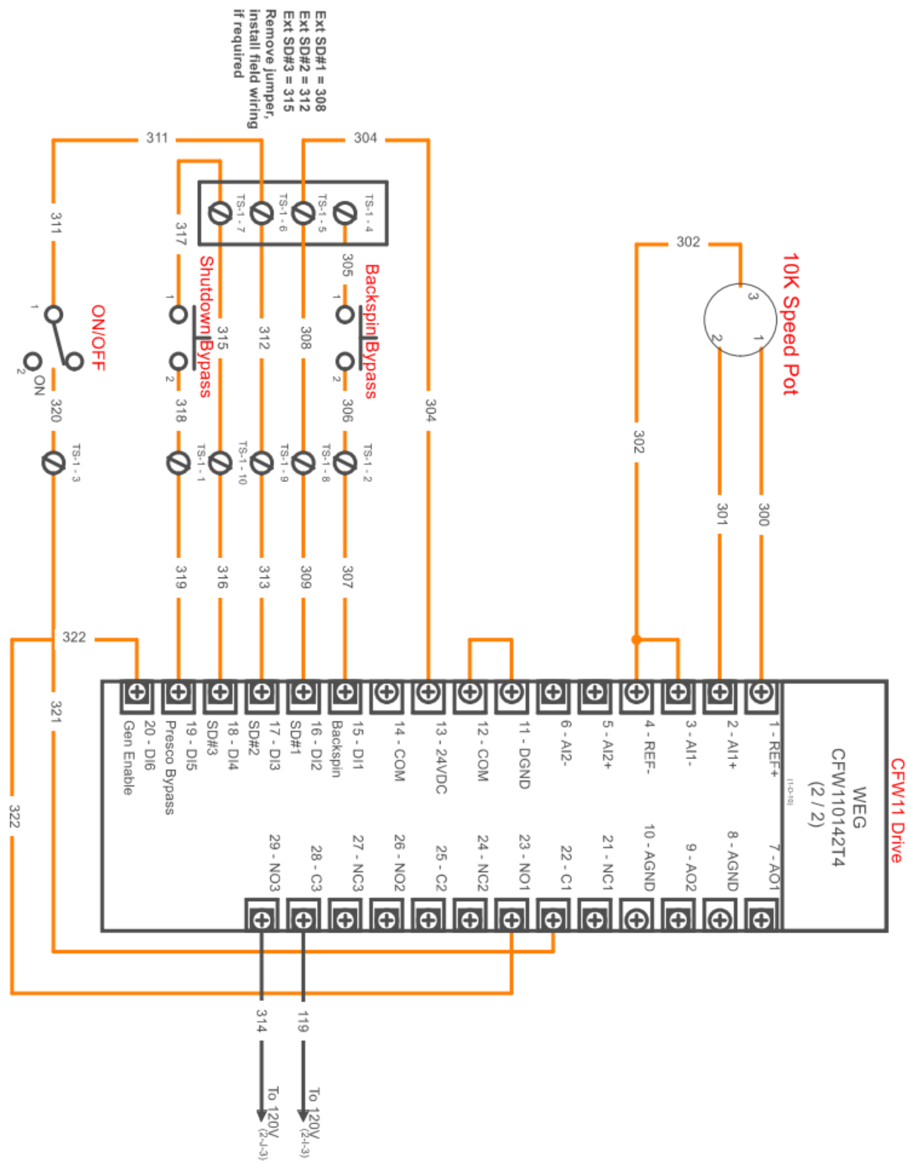
For common faults/alarms and their possible causes see table starting on page 13.

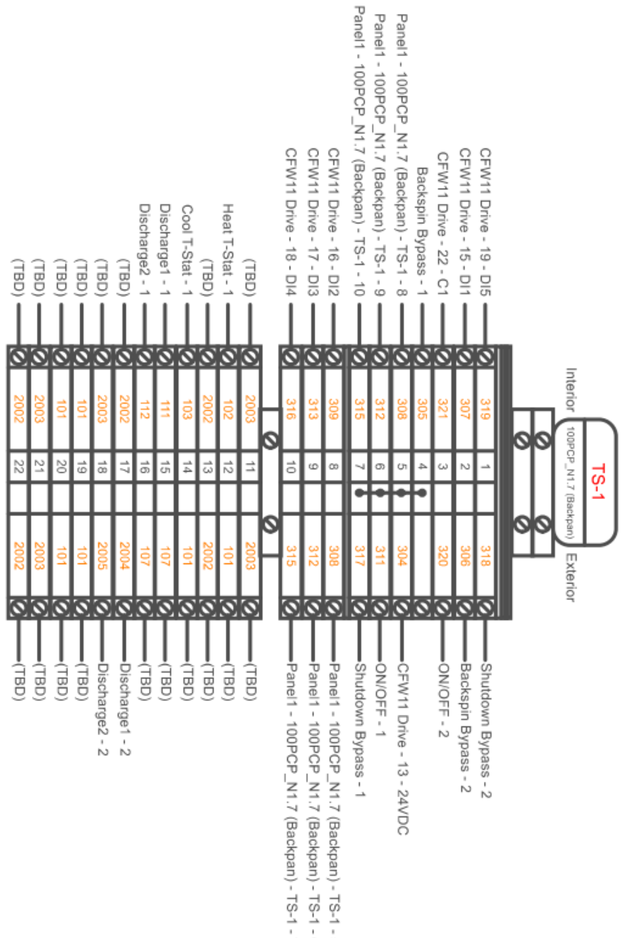
Faults/Alarms	Description	Possible Causes
F006 Imbalance or Input Phase Loss	Mains voltage imbalance too high or phase missing in the input power supply. Note: - If the motor is unloaded or operating with reduced load this fault may not occur. - Fault delay is set at parameter P0357. P0357 = 0 disables the fault. - If single-phase power supply is used, this fault must be disabled.	Phase missing at the inverter's input power supply. Input voltage imbalance >5 %. For the Frame Size E: Phase loss at L3/R or L3/S may cause F021 or F185. Phase loss at L3/T will cause F006. For frame sizes F and G: Pre-charge circuit fault.
F021 DC Link Undervoltage	DC Link undervoltage condition occurred	The input voltage is too low, and the DC bus voltage dropped below the minimum permitted value (monitor the value at parameter P0004): DC BUS Voltagea < 446 V - For a 440-460 V input voltage (P0296 = 3). DC BUS Voltage < 487 V - For a 480 V input voltage (P0296 = 4). Phase loss in the input power supply. Pre-charge circuit failure. Parameter P0296 was set to a value above of the power supply rated voltage

F022 DC Link Overvoltage	DC link overvoltage condition occurred.	The input voltage is too high, and the DC bus voltage surpassed the maximum permitted value: DC Bus Voltage > 400 V - For 220-230 V input models (P0296 = 0). DC Bus Voltage > 800 V - For 380-480 V input Inertia of the driven load is too high, or deceleration time is too short. Wrong settings for parameters P0151, or P0153, or P0185. **Ensure P0185 and P0186 are set to 770 and 40 respectively.**
F048 IGBT Overload Fault	An IGBTs overload fault occurred.	Inverter output current is too high.
F070 Overcurrent / Short-circuit	Overcurrent or short-circuit detected at the output, in the DC bus, or at the braking resistor.	Short-circuit between two motor phases. Short-circuit between the connection cables of the dynamic braking resistor. IGBT modules are shorted. Disconnect Motor try running VFD if issue persists replace inverter, if problem goes away check motor and cable for shorts perform a megger test.
F071 Output Overcurrent	The inverter output current was too high for too long.	Excessive load inertia or acceleration time too short. Settings of P0135 or P0169 and P0170 are too high
F072 Motor Overload	The motor overload protection operated. Note: It may be disabled by setting P0348 = 0 or 3	Settings of P0156, P0157 and P0158 are too low for the used motor. Motor shaft load is excessive.
F0150 Motor Overspeed	Overspeed fault. It is activated when the real speed exceeds the value of P0134 x (100 % + P0132) for more than 20 ms.	Wrong settings of P0161 and/or P0162. Problem with the hoist-type load.

A163/164 Break Detect AI1/AI2	It indicates that the AI current (4-20 mA or 20-4 mA) reference is out of the 4 to 20 mA range.	Broken AI cable. Bad contact at the signal connection to the terminal strip. Blown control fuse on terminal strip for AI circuit. If there is no AI going to that input set the corresponding input **P0233-AI1 or P0238-AI2 to 0.**
A181 Invalid Clock Value	Invalid clock value alarm	It is necessary to set date and time at parameters P0194 to P0199. (Try this first if not then check battery) Keypad battery is discharged, defective, or not installed.
F185 Pre-charge Contactor Fault	It indicates fault at the pre-charge contactor	Pre-charge contactor defect. Open command fuse. Phase loss in the input L1/R or L2/S. P0355 = 1 (Incorrect setting for model's frame E supplied by the DC link. For those models, the setting should be P0355 = 0.
A750 External Shutdown #1	Indicates an external shutdown is tripped	Check Presco, check vibration switch, check pollution switch, ensure jumpers left are tight and secure. Ensure 24VDC loop is complete.
A751 External Shutdown #2	Indicates an external shutdown is tripped	Check Presco, check vibration switch, check pollution switch, ensure jumpers left are tight and secure. Ensure 24VDC loop is complete.
A752 External Shutdown #3	Indicates an external shutdown is tripped	Check Presco, check vibration switch, check pollution switch, ensure jumpers left are tight and secure. Ensure 24VDC loop is complete.

“Close General Enable” won’t go away and drive wont autotune during setup	Soft PLC is not set to run program or loose/incorrect wiring	Ensure Soft PLC Command P1001 is set to “Run Program” Verify wiring is correct and connections are tight. Press Green button, ensure on/off switch is in the on position Ensure all S/D cables are connected and not tripped. Switch parameter P0202 back to V/F 60hz and ensure “Ready is displayed in top left-hand corner”
VFD wont run after power bump/loss	When the on/off switch is on, but the VFD does not run	Ensure that the green HMI button has been pressed. Alternatively, if you want the VFD to restart automatically Parameter P1035 can be enabled for power loss restart
VFD stuck in auto tuning menu	Auto tune not completed	Ensure auto tune has been completed scroll down to bottom to P0408 to “Run self tuning?” set to 1 for “No Rotation” turn hand switch on and press green button
VFD going down on A750	High Torque Shutdown	VFD is tripping on high torque shutdown, verify shutdown has been set and the value is correct. Verify polish rod torque on screen isn’t rising above value on main screen of HMI


CFW11 Drive
WEG
CFW11014 2T4
(2 / 2)
10K Speed Pot
Backspin Bypass
Shutdown Bypass
ON/OFF
ON
Ext SD#1 = 308
Ext SD#2 = 312
Ext SD#3 = 315
Remove jumper,
install field wiring
if required
1 - REF+
2 - AI1+
3 - AI1-
4 - REF-
5 - AI2+
6 - AI2-
11 - DGND
12 - COM
13 - 24VDC
14 - COM
15 - DI1
Backspin
16 - DI2
SD#1
17 - DI3
SD#2
18 - DI4
SD#3
19 - DI5
Presco Bypass
20 - DI6
Gen Enable
7 - AO1
8 - AGND
9 - AO2
10 - AGND
21 - NC1
22 - C1
23 - NO1
24 - NC2
25 - C2
26 - NO2
27 - NC3
28 - C3
29 - NO3
To 120V (2-J-3)
To 120V (2-I-3)


TS-1
100PCP_N1.7 (Backpan)
Interior
Exterior
CFW11 Drive - 19 - DI5
CFW11 Drive - 15 - DI1
CFW11 Drive - 22 - C1
Backspin Bypass - 1
Panel1 - 100PCP_N1.7 (Backpan) - TS-1 - 8
Panel1 - 100PCP_N1.7 (Backpan) - TS-1 - 9
Panel1 - 100PCP_N1.7 (Backpan) - TS-1 - 10
CFW11 Drive - 16 - DI2
CFW11 Drive - 17 - DI3
CFW11 Drive - 18 - DI4
(TBD)
Heat T-Stat - 1
(TBD)
Cool T-Stat - 1
Discharge1 - 1
Discharge2 - 1
(TBD)
(TBD)
(TBD)
(TBD)
(TBD)
(TBD)
319 307 321 305 308 312 315 309 313 316 2003 102 2002 103 111 112 2002 2003 101 101 2003 2002
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22
318 306 320 304 311 317 308 312 315 2003 101 2002 101 107 107 2004 2005 101 101 2003 2002
Shutdown Bypass - 2
Backspin Bypass - 2
ON/OFF - 2
CFW11 Drive - 13 - 24VDC
ON/OFF - 1
Shutdown Bypass - 1
Panel1 - 100PCP_N1.7 (Backpan) - TS-1 - 5
Panel1 - 100PCP_N1.7 (Backpan) - TS-1 - 6
Panel1 - 100PCP_N1.7 (Backpan) - TS-1 - 7
(TBD)
(TBD)
(TBD)
(TBD)
(TBD)
(TBD)
Discharge1 - 2
Discharge2 - 2
(TBD)
(TBD)
(TBD)
(TBD)

For CFW11 Alarm & Fault List refer to page 6-2 of the CFW-11 User Manual Provided

SEEL Software Alarm List:

A750 – High Torque Shutdown

A751 – Backspin

A752 – Shutdown #1 (Auto Restart can be enabled)

A753 – Shutdown #2 (Does not have auto restart capability)

A754 – Shutdown #3 (Does not have auto restart capability)

A755 – Auto Restart Wait Time Delay

Analog Input / Output Configuration: (Dipswitch's Located at top of control board)

S1.1 (Analog Output 1): Dip Switch Off - 4-20mA, Dip Switch On: 0-10vdc

S1.2 (Analog Output 2): Dip Switch Off - 4-20A, Dip Switch On: 0-10vdc

S1.3 (Analog Input 2): Dip Switch Off – 0-10vdc, Dip Switch On: 4-20mA

S1.4 (Analog Input 1): Dip Switch Off – 0-10vdc, Dip Switch On: 4-20mA

Notes:

- Red Button on Keypad is utilized to reset alarms/faults, this will also reset the on/off timer.
- Green Button on Keypad is utilized to start VFD in local.
- If equipped red beacon illuminates when shutdown #1, # 2, 3 are tripped or when a VFD alarm or fault has occurred.

Digital Inputs:

- DI1 – Backspin Bypass Push Button
- DI2 – Shutdown #1 Connection (Normally Closed), this shutdown can be programmed for auto restart
- DI3 – Shutdown #2 Connection (Normally Closed), this shutdown does not have auto restart capability
- DI4 – Shutdown #3 Connection (Normally Closed), this shutdown does not have auto restart capability
- DI5 – Shutdown Bypass Push Button (for testing shutdowns while unit stays running)
- DI6 - General Enable (Normally Open)

Technical Support

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Notes: